

Simplifying fractions

While TSM 099 studies Topic 6. Review: Topic 4.

Equivalent forms are acceptable. Use these answers after attempting the problems; revisit the linked lecture notes for additional examples.

Core: simplifying fractions

1. Write in lowest terms.

$$\frac{18}{24} = \frac{3}{4}$$

2. Write in lowest terms.

$$\frac{21}{35} = \frac{3}{5}$$

3. Write in lowest terms.

$$\frac{28}{42} = \frac{2}{3}$$

4. Write in lowest terms.

$$\frac{45}{60} = \frac{3}{4}$$

5. Write in lowest terms.

$$\frac{48}{72} = \frac{2}{3}$$

6. Write in lowest terms.

$$\frac{54}{90} = \frac{3}{5}$$

7. Write in lowest terms.

$$\frac{64}{80} = \frac{4}{5}$$

8. Write in lowest terms.

$$\frac{84}{126} = \frac{2}{3}$$

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Core: signs and further reduction

9. Write in lowest terms.

$$\frac{-24}{36} = -\frac{2}{3}$$

10. Write in lowest terms.

$$\frac{30}{-45} = -\frac{2}{3}$$

11. Write in lowest terms.

$$\frac{-42}{-56} = \frac{3}{4}$$

12. Write in lowest terms.

$$\frac{0}{19} = 0$$

13. Write in lowest terms.

$$\frac{96}{144} = \frac{2}{3}$$

14. Write in lowest terms.

$$\frac{150}{210} = \frac{5}{7}$$

15. Write in lowest terms.

$$\frac{81}{54} = \frac{3}{2}$$

16. Write in lowest terms.

$$\frac{128}{160} = \frac{4}{5}$$

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Core: equivalent fractions

17. Find the missing numerator.

? = 8
Multiply numerator and denominator by 4.

18. Find the missing numerator.

? = 18
Multiply numerator and denominator by 6.

19. Find the missing numerator.

? = -12
Multiply numerator and denominator by 3.

20. Find the missing numerator.

? = 35
Multiply numerator and denominator by 7.

21. Find the missing numerator.

? = 28
Multiply numerator and denominator by 4.

22. Find the missing numerator.

? = 0
Multiply numerator and denominator by 6.

23. Find the missing numerator.

? = 55
Multiply numerator and denominator by 5.

24. Find the missing numerator.

? = -24
Multiply numerator and denominator by 8.

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Extension: compare and explain

25. Are these fractions equivalent? Justify.

$$\frac{3}{4} \text{ and } \frac{3}{4}$$

Yes; the reduced values match.

26. Are these fractions equivalent? Justify.

$$\frac{4}{9} \text{ and } \frac{8}{15}$$

No; the reduced values differ.

27. Are these fractions equivalent? Justify.

$$-\frac{3}{5} \text{ and } -\frac{3}{5}$$

Yes; the reduced values match.

28. Are these fractions equivalent? Justify.

$$\frac{2}{3} \text{ and } \frac{3}{4}$$

No; the reduced values differ.

29. Write in lowest terms.

$$\frac{252}{420} = \frac{3}{5}$$

30. Write in lowest terms.

$$\frac{-165}{225} = -\frac{11}{15}$$

31. Explain why dividing only the numerator by 3 changes the value.

$$\frac{9}{12} \neq \frac{3}{12}$$

The original is $\frac{3}{4}$; the new fraction is $\frac{1}{4}$. Divide both parts by the same nonzero factor to preserve value.

32. Can multiplying both numerator and denominator by zero produce an equivalent fraction? Explain.

No. It produces $0/0$, which is undefined.